[ ]:
```
Apply prompting to semi-automate a useful real-world task such as summarizing notes, extracting structured data, or transforming.
Design a clear, consistent prompt that can handle multiple inputs and produce reliable output.
```

[1]:
```
# ============================================================================
# CELL 1: Setup
# ============================================================================

print("Project: Automated Summarization & Data Extraction")
```

```
Project: Automated Summarization & Data Extraction
```

[2]:
```
# ============================================================================
# CELL 2: Prompt Design
# ============================================================================

prompt_template = """
Summarize the given text and extract key information.

Output format:

Summary:
(2-3 lines)

Key Points:
- Point 1
- Point 2
- Point 3

JSON:
{
  "topic": "",
  "main_points": [],
  "keywords": []
```

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Code JupyterLab Python 3 (ipykernel)

```
prompt_template = """
Summarize the given text and extract key information.

Output format:

Summary:
(2-3 lines)

Key Points:
- Point 1
- Point 2
- Point 3

JSON:
{
  "topic": "",
  "main_points": [],
  "keywords": []
}

Rules:
- Keep it short and clear
- Do not add extra text
- Follow format strictly

Text:
"""

print("Prompt template created successfully")
```

```
Prompt template created successfully
```

[3]:

```
# ============================================================================
# CELL 3: Inputs
# ============================================================================
```

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Code JupyterLab Python 3 (ipykernel)

[3]:
```python
# ==============================================================================
# CELL 3: Inputs
# ==============================================================================

input1 = "Artificial Intelligence helps machines learn from data and improves efficiency in healthcare and finance."

input2 = "Cloud computing allows storing and accessing data over the internet with flexibility and scalability."

input3 = "Machine learning is a part of AI that identifies patterns in data and is used in recommendation systems."

print("Inputs ready")
```

```
Inputs ready
```

[4]:
```python
# ==============================================================================
# CELL 4: Outputs
# ==============================================================================

output1 = """
Summary:
AI helps machines learn from data and improves efficiency.

Key Points:
- Learns from data
- Used in healthcare and finance
- Improves efficiency

JSON:
{
  "topic": "Artificial Intelligence",
  "main_points": ["Learning from data", "Industry applications", "Efficiency improvement"],
  "keywords": ["AI", "data", "automation"]
}
"""
```

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Code

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```
output2 = """
Summary:
Cloud computing enables online data storage with scalable solutions.

Key Points:
- Data stored online
- Flexible access
- Scalable systems

JSON:
{
  "topic": "Cloud Computing",
  "main_points": ["Online storage", "Flexible access", "Scalability"],
  "keywords": ["cloud", "storage", "internet"]
}
"""

output3 = """
Summary:
Machine learning finds patterns in data and supports smart systems.

Key Points:
- Subset of AI
- Pattern recognition
- Used in recommendations

JSON:
{
  "topic": "Machine Learning",
  "main_points": ["Part of AI", "Pattern detection", "Recommendation systems"],
  "keywords": ["ML", "AI", "data"]
}
"""
```

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Code JupyterLab Python 3 (ipykernel)

```
  "keywords": ["ML", "AI", "data"]
}
"""

print("Outputs generated")
```

```
Outputs generated
```

[5]:

```
# ============================================================================
# CELL 5: Display Results
# ============================================================================

print("\nINPUT 1:\n", input1)
print("\nOUTPUT 1:\n", output1)

print("\nINPUT 2:\n", input2)
print("\nOUTPUT 2:\n", output2)

print("\nINPUT 3:\n", input3)
print("\nOUTPUT 3:\n", output3)
```

```
INPUT 1:
 Artificial Intelligence helps machines learn from data and improves efficiency in healthcare and finance.

OUTPUT 1:

Summary:
AI helps machines learn from data and improves efficiency.

Key Points:
- Learns from data
- Used in healthcare and finance
- Improves efficiency

JSON:
{
```

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Code JupyterLab Python 3 (ipykernel)

```
JSON:
{
  "topic": "Artificial Intelligence",
  "main_points": ["Learning from data", "Industry applications", "Efficiency improvement"],
  "keywords": ["AI", "data", "automation"]
}


INPUT 2:
 Cloud computing allows storing and accessing data over the internet with flexibility and scalability.

OUTPUT 2:

Summary:
Cloud computing enables online data storage with scalable solutions.

Key Points:
- Data stored online
- Flexible access
- Scalable systems

JSON:
{
  "topic": "Cloud Computing",
  "main_points": ["Online storage", "Flexible access", "Scalability"],
  "keywords": ["cloud", "storage", "internet"]
}


INPUT 3:
 Machine learning is a part of AI that identifies patterns in data and is used in recommendation systems.

OUTPUT 3:

Summary:
Machine learning finds patterns in data and supports smart systems.
```

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Code JupyterLab Python 3 (ipykernel)

```
Key Points:
- Subset of AI
- Pattern recognition
- Used in recommendations

JSON:
{
  "topic": "Machine Learning",
  "main_points": ["Part of AI", "Pattern detection", "Recommendation systems"],
  "keywords": ["ML", "AI", "data"]
}
```

[6]:

```python
# ============================================================================
# CELL 6: Reflection
# ============================================================================

print("REFLECTION:\n")

print("Initial Issues:")
print("- Output format was inconsistent")
print("- JSON was sometimes missing")
print("- Extra text was added\n")

print("Improvements Made:")
print("- Added strict format instructions")
print("- Defined clear sections (Summary, Key Points, JSON)")
print("- Added rules to avoid extra text\n")

print("Final Result:")
print("- Consistent outputs")
print("- Works for different inputs")
print("- Useful for real-world automation tasks")
```

```
REFLECTION:
```

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Code JupyterLab Python 3 (ipykernel)

```python
print("REFLECTION:\n")

print("Initial Issues:")
print("- Output format was inconsistent")
print("- JSON was sometimes missing")
print("- Extra text was added\n")

print("Improvements Made:")
print("- Added strict format instructions")
print("- Defined clear sections (Summary, Key Points, JSON)")
print("- Added rules to avoid extra text\n")

print("Final Result:")
print("- Consistent outputs")
print("- Works for different inputs")
print("- Useful for real-world automation tasks")
```

```
REFLECTION:

Initial Issues:
- Output format was inconsistent
- JSON was sometimes missing
- Extra text was added

Improvements Made:
- Added strict format instructions
- Defined clear sections (Summary, Key Points, JSON)
- Added rules to avoid extra text

Final Result:
- Consistent outputs
- Works for different inputs
- Useful for real-world automation tasks
```

[ ]: